

FALL 2019 | M&W 10:05-11:20am | Room: Old Chem 123

POLISCI 748: INTRO TO CAUSAL INFERENCE

Companion course to POLISCI 633: Intro to Theory

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“Thus we remember to have seen that species of object we call *flame*, and to have felt that species of sensation we call *heat*. We likewise call to mind their constant conjunction in all past instances. Without any farther ceremony, we call the one *cause* and the other *effect*, and infer the existence of the one from that of the other.”

— David Hume, 1896

“We think of a cause as something that makes a difference, and the difference it makes must be a difference from what would have happened without it. Had it been absent, its effects — some of them, at least, and usually all — would have been absent as well.”

— David Lewis, 1973

Course Overview

This course is a graduate-level introduction to the logic and practice of causal inference. Causal inference is the process of drawing (or trying to draw, as the case may be) causal conclusions from evidence. The course will provide you with a non-technical — that is, a logical as opposed to a formal mathematical — overview of different approaches to drawing such causal conclusions. The

course is designed to present to you the core logic of the comparisons made in different standard research designs aimed at causal inference, and critically, the assumptions that must be true in order for those comparisons to generate true causal results. Our course will not exhaust the set of research designs that are available to empirical scholars, but will provide you firm grounding in the most used.

From this logical background, students should be well-positioned to seek out the relevant statistical models that allow them to implement those research designs when they are quantitative, recognize how those statistical models map onto the causal models, and how statistical inference relates to causal inference. But because causal inference is *not* a quantitative practice, but an evidentiary practice broadly, students will also be well-positioned to engage with standard qualitative models of causal inference, and utilize qualitative evidence to support quantitative research designs.

In addition to the focus on causal inference, the course is also designed to help students to develop their own empirical research designs to answer causal questions. Therefore we will both study the logic of causal inference in the abstract and in the work of others, and *apply* that logic to student research questions. By the end of the semester, students will have produced a short research proposal that outlines a question, a theory in answer to that question, a hypothesis (or *-ses*) to test that theory, a measurement strategy, a feasible research design to evaluate the hypothesis, a description of the assumptions necessary to interpret the design as causal, and a plan for providing the evidence necessary to support those assumptions.

These proposals may form the basis for a future paper, or end up in the bottom of a desk drawer never again to see the light of day; however, the *practice* developing these ideas, regardless of the outcome, will be the critical component. Scholars rarely hit on a good design their first go-round. It takes time and practice and lots of failure. And without trying, you cannot succeed.

Big Ideas of the Course

A few big ideas and enduring understandings will facilitate our exploration of causal inference in this course. The first big idea is that *causality is fundamentally unknowable, and we must therefore approximate an unobserved and unobservable outcome* in order to draw causal inferences. This is sometimes referred to as the fundamental problem of causal inference — the idea that an

understanding of cause and effect depends on our ability to understand outcomes that never happened. In this way, I (personally) think the study and application of causality is about as close as science gets to magical thinking, which, when you think about it deeply, is pretty cool. Thus, our work as empirical social scientists is in how we *approximate* the unobserved counterfactual outcome, with the understanding that it will *always be* an approximation.

The second big idea is that *causal inference is a logical process of making comparisons, not a quantitative nor a qualitative process*. Regardless of which type of evidence you are utilizing, causality depends on making thoughtful comparisons that allow you the best opportunity to approximate an unobservable occurrence. In understanding what this means, we will hopefully dispel (or ward off) notions that one type of research is inherently better than another, and see that the choice of evidentiary approach, and research design more broadly, are question-driven rather than ideological.

Finally, the third big idea is that *causal inference always requires assumptions, and evaluating those assumptions requires judgement*. The extent to which we think the necessary assumptions are *true* will dictate how confident we are in a given causal inference. But we will always need assumptions, because our approaches are only approximating the counterfactual. Yet evaluating those assumptions is an inherently subjective process — there is no test, no summary statistic, no one number or set of numbers that will tell you exactly whether an assumption is true or not.¹ And although some of the assumptions we'll study have statistical analogues, we'll learn the importance of recognizing (in the vein of the big idea above) that assumptions for causal inference are logical assumptions, not necessarily statistical assumptions.

Companion Course

This course is paired with POLISCI 633 to provide you an introduction *both* to the process of theory-building and hypothesis generation in political science, as well as the design of research for causal inference. These courses are har-

¹As an aside, this is the same as statistical inference in the frequentist tradition of statistics. The choice, for instance, of which *p-value* one is willing to accept in order to believe that an estimate of a parameter is different from some null hypothesis value is a personal, subjective choice. It is a choice that balances the probability of two types of opposing errors — Type I and Type II. While there may be *norms* that see groups converge on being willing to accept a particular value — say, 5% — there is no objective truth that makes that choice of a value “better” than another.

monized, although still distinct, and culminate in student-developed research proposals which are presented in joint class sessions.

Because of these presentations, this course is a short course. The first half of the semester includes lectures and discussions, while the second half of the semester is dedicated entirely to student presentations.

Course Structure

The course is structured as a seminar in which discussing readings, asking questions, and probing topics of interest will be the key components. In addition, class time will also be used to workshop components of students' original research proposals.

Readings

There is no textbook for the course. Instead, we will read papers and selected chapters from a variety of books. There will also be one or two additional media that are classified as "readings," most notably podcasts. **All readings will be available on the course website on Sakai.** Readings will be under the relevant week's folder, labeled with the week (W1, W2, W3, etc.), labeled with the the class (C1 is class 1 on Monday, and C2 is class 2 on Wednesday), and the author names. You are expected to complete the readings listed for a given class *before* that day's class.

Learning Goals

By the end of the course, you will be able to

1. Differentiate between *descriptive* research questions and *causal* research questions; and identify a related causal question to a given descriptive question (and vice versa).
2. Articulate in your own words the fundamental problem of causal inference.
3. Describe both the counterfactual condition for a given causal question

and the ideal (albeit potentially infeasible) experiment that would allow for the best causal answer to a given question.

4. Identify the core comparisons made in different standard research designs, how each design approximates the counterfactual condition, and the assumptions that must be true for the comparison to generate a causal result.
5. Write a succinct research proposal and present it in a short-format conference environment in a way that allows your colleagues the ability understand your plan and offer you constructive feedback.
6. Prepare and deliver a short discussion of your colleague's work that highlights a select few points for improvement, and crucially emphasizes *constructive thoughtful* suggestions for how to make those improvements and develop the research project further.

Assignments

A core goal of this course is to give you the opportunity at the very outset of your graduate career to design your own research to answer questions about the political world that are interesting to you. In this course, that entails the application of your knowledge about what causality is and how counterfactuals can be approximated to particular questions that interest you.

The following assignments are designed to help you to achieve the above learning goals, thereby deepening your abstract understanding of causal inference, and developing your ability to apply that abstract understanding to your own research. More information about each will be provided in class.

Three Research Questions

DUE: Wednesday, September 11, in class

The first assignment is a research question assignment. You will submit three distinct causal research questions. You do not need to write a substantial amount. Instead you just need to come with three research questions that you can discuss and workshop with your fellow students. In class, you will work on refining the questions. You'll discuss the counterfactual to each, and what

the experimental ideal would be.

Two Research Designs

DUE: Wednesday, October 2, in class

The second assignment asks you to select one of your research questions from the assignment above (or a new question, or the revised version of a question post-in-class workshop). You will then present two *different* research designs that can potentially answer the question. These should be feasible designs. In class, you will discuss your designs and workshop them with students. In particular, you'll work on laying out the strengths of the designs, the potential threads to inference, and discussing how you might support the assumptions necessary for interpreting each design as causal.

Draft Research Proposal

DUE: Three days before your in-class presentation, by email to me and your discussant

The draft research proposal asks you to pull together information from your above assignments into one **two-page** proposal. Given the very short length of the document, the goal with this proposal is to be concise and utilitarian. Your draft proposal should present your (1) research question, present a (2) sketch of the theory and hypotheses you develop in POLISCI 633, explain the (3) ideal experiment to answer your research question, even if infeasible, the (4) feasible (if not ideal) research design you've chosen, a description of (5) the logical assumptions for making a causal inference from that design, and (6) how you plan to substantiate those assumptions.

In-Class Presentation

DUE: To be determined

You will present your proposal in a five minute conference-style presentation with slides. This presentation will occur in the second half of class. You will see a model presentation in the style we expect prior to undertaking your own presentation.

In-Class Discussant

DUE: To be determined

You will serve as the in-class discussant for the presentation of one of your peers. This will be extremely short. You will present for just two minutes. You'll be expected to offer feedback on what is working well in the presentation, and concrete constructive suggestions for what could be improved. You will

see a model of the discussion we expect prior to undertaking it on your own.

Final Research Proposal

DUE: Sunday, December 8, by the end of the day, by email

The final research proposal incorporates the feedback that you've received on your two page draft proposal from your instructor, fellow students and discussant. You should demonstrate meaningful steps improving your proposal relative to the draft you submitted.

Participation

The success of the course will depend on the active engagement of all students in class. I expect all students not only to have completed readings prior to class, but to have thought about them and reflected on them. Knowing the quantity of work you have for other classes, I am keeping the reading load very light for this class. But in unburdening you from a large quantity of reading, I nevertheless expect your engagement with that reading to be of the highest quality.

While you will develop (or refine) your own strategies of being prepared to participate in class, two of the best strategies are to write a set of questions and thoughts about the reading that you are interested in sharing with the group. Although it can be hard to be vulnerable, I deeply encourage students to bring questions to class of the variety "I don't understand *this*" or "I thought *this*, but this reading confused it" — that is, questions that reveal uncertainty. These are typically the questions that allow *everyone* in class the opportunity to clarify and solidify their understanding.

Informal assignments that ask you to do short exercises in relationship to readings will be part of your participation grade.

Grading

Letter grades in a PhD program are not particularly important. As a doctoral student, you are beginning your own career as an independent scholar, and that independence means that you are in charge of your learning in a way that you (likely) never have been before. My job as your instructor is to provide

you with feedback on your development as a scholar, and more specifically, your progress meeting the above-stated learning goals. As a consequence, the content of the feedback that I give you is far more important as you develop your ideas throughout this course than the particular letter grade or mark that you receive.

Therefore, assignments will be graded on the following simple basis:

✓+ Indicates outstanding work that goes above and beyond the requirements for the assignment. Demonstrates deeper understanding than expected, a profound innovation, or exemplary effort.

✓ Indicates work that is well done and which meets the requirements for the assignment. You will probably receive this mark most of the time.

✓- Indicates work that evidences gaps in understanding of the material. You will not receive this mark without extensive explanation of where I think you are struggling to meet the learning goal. If you receive this mark, we will likely meet in person to discuss gaps in understanding and collaborate together on filling them.

Because I will need to enter final grades into the system that are of the A, B, C variety, you should expect to receive an A/A- if you've received mostly ✓. If you have some ✓+ it's possible to receive an A+. Students with some ✓- may receive a B+, or in extreme cases a B. Receiving lower than a B indicates that something extreme has occurred — for instance, you did not attend class, or you did not complete assignments.

With all marks — whether on individual assignments or in the course overall — I strongly encourage you take a growth mindset. A growth mindset requires that you believe that your abilities are not fixed based on some immutable notion of intelligence or talent, but are instead under your control to develop and improve through dedication and hard work. If you receive a top mark, this doesn't mean that there isn't more to do and more to learn. Similarly, if you receive a low mark, this is not a referendum on your intelligence or your potential. Instead, it is constructive feedback that is designed to help you to do better in the future.

Laptop Policy

The causal evidence from the teaching and learning field clearly shows that we

retain information better when we write with paper and a pen, rather than type our notes. This is primarily a function of the information synthesis that has to occur when we write because we are not fast enough writers to transcribe everything being said. Most typists, on the other hand, can conceivably transcribe word for word for what's being said, but this requires little to no mental processing of the information.

For this reason, I strongly prefer that you take written notes and refrain from using your laptop in class.

Resources

Appointments with Me

Please reach out to make an appointment with me if I can be helpful with your research proposal or your understanding of the class material. If you have a brief question, you can ask over email as well. It may take me up to 24 hours to respond, however, I will absolutely respond.

Your Fellow Students

Academic scientific is a highly collaborative process. Even when this is not evidenced in formal co-authorships, the way in which your peers can probe, push, and refine your ideas is one of the most valuable (and rewarding, I think) parts of our discipline. Thus, you are welcome to share your ideas widely with students in your cohort or beyond, incorporate their thoughts, and rely on their advice. This will undoubtedly make your final project better.

At the same time, for other small assignments or in approaching the readings, I encourage you to work independently *first*, before having discussions with your fellow students. This will give you an opportunity to practice applying the knowledge that we're developing in class, and gauge your own understanding of the material. However, once you've tried something on your own, seek out your fellow students to share. We're smarter together.

Mental Health

Graduate school is designed to stretch your intellectual abilities. Doing so necessitates criticism (though it should *always* be of the constructive variety).

For many students this is some of the first criticism that they've ever received. Graduate school also puts you in charge of your own scholarship in a way few students have previously experienced. That freedom can be powerful, but also overwhelming. Finally, the sheer amount of work that you're asked to do can leave you eating poorly, sleeping poorly, and neglecting yourself in other basic ways.

All of this stress means that taking care of your mental health is a critical component of graduate school. Learning skills to manage stress, anxiety, and disappointment will make you happier in life overall and a more successful scholar. For some of you, taking care of your mental health will require few adjustments from before – prioritizing sleep, taking vacation days to get away from the stress, or actively cultivating your interests outside of grad school to keep healthy life balance. For others of you, you may want to invest more in mental health. In particular, enlisting the help of a professional counselor to teach your mental health skills can be invaluable. The Duke Wellness Center is a great resource for this.

SCHEDULE

PART I

Week 1

Class 1 — August 26 — Causality in the Time of Cholera

Class 2 — August 28 — Descriptive and Causal Questions

1. Gerring, John (2012). “Mere Description.” *British Journal of Political Science*.
2. Grimmer, Justin (2013). *Representational Style: What Legislators Say and Why it Matters*. New York: Cambridge University Press. **Chapters 3, 4 and 5.**

Week 2

Class 1 — September 2 — Counterfactuals and the Potential Outcomes Framework

1. Morgan, Stephen L. and Christopher Winship (2015). *Counterfactuals and Causal Inference*. Second Edition. New York: Cambridge University Press. **Pages 37-58.**

Class 2 — September 4 — Mill’s Methods

1. Sekhon, Jasjeet S. (2008). “The Neyman-Rubin Model of Causal Inference and Matching.” in *The Oxford Handbook of Political Methodology*. Janet M. Box-Steffensmeier, Henry E Brady and David Collier, eds. New York: Oxford University Press. **Pages 281-299.**
2. Lindsey, Carrie Lee (2015). *The Politics of Military Operations*. Dissertation. **Parts 1 and 2.**

Week 3

Class 1 — September 9 — The Experimental Ideal

1. Angrist, Joshua D. and Jorn-Steffan Pischke (2009). *Mostly Harmless Econometrics*. Princeton, NJ: Princeton University Press. **Pages 3-24.**
2. Gottlieb, Jessica (2016). “Greater Expectations: A Field Experiment to Improve Accountability in Mali.” *American Journal of Political Science*. Vol. 60, No. 1.

Class 2 — September 11 — Research Question Workshop

1. No readings. Three research questions assignment due.

Week 4

Class 1 — September 16 — (True) Natural Experiments

1. Dunning, Thad (2008). Improving Causal Inference: Strengths and Limitations of Natural Experiments. *Political Research Quarterly*. Vol. 61. No. 2.
2. Hyde, Susan (2007). “The Observer Effect in International Politics: Evidence from a Natural Experiment.” *World Politics*. Vol. 60, No. 1.
3. 99% Invisible. “Episode 352: Uptown Squirrel.” URL: <https://99percentinvisible.org/episode/uptown-squirrel>

Class 2 — September 18 — Matching

1. Stuart, Elizabeth (2010). “Matching Methods for Causal Inference: A Review and a Look Forward.” *Statistical Science*.
2. Lyall, Jason (2010). “Are Coethnics More Effective Counterinsurgents? Evidence from the second Chechen War.” *American Political Science Review*. Vol. 104, No. 1.

Week 5

Class 1 — September 23 — NO CLASS

Class 2 — September 25 — Regression Discontinuity Design

1. Angrist, Joshua D. and Jorn-Steffan Pischke (2009). *Mostly Harmless Econometrics*. Princeton, NJ: Princeton University Press. **Pages 249-268.**
2. Ferwerda, Jeremy and Nicholas L. Miller (2014). “Political Devolution and Resistance to Foreign Rule: A Natural Experiment.” *American Political Science Review*.
3. Kocher, Matthew A. and Nuno P. Monteiro. “Lines of Demarcation: Causation, Design-Based Inference, and Historical Research.” *Perspectives on Politics*.

Week 6

Class 1 — September 30 — Difference-in-Differences

- (a) Payson, Julia (2018). “Cities in the Statehouse: How Local Governments Use Lobbyists to Secure State Funding.” *Working Paper*.
- (b) Angrist, Joshua D. and Jorn-Steffan Pischke (2009). *Mostly Harmless Econometrics*. Princeton, NJ: Princeton University Press. **Pages 222-247.**

Class 2 — October 2 — Research Design Proposal Workshop

- (a) No readings. Two proposal research designs are due.

Week 7

Class 1 — October 7 — NO CLASS FALL BREAK

Class 2 — October 9 — NO CLASS FALL BREAK

Week 8

Class 1 — October 14 — Instrumental Variables

1. Angrist, Joshua D. and Jorn-Steffan Pischke (2009). *Mostly Harmless Econometrics*. Princeton, NJ: Princeton University Press. **Pages 114-218.**
2. Acemoglu, Daron, Simon Johnson, and James A. Robinson. “Colonial Origins of Comparative Development: An Empirical Investigation.” *American Economic Review*.

Class 2 — October 16 — Faculty Model Presentations and Discussion

1. No readings. Two faculty will model presentation formats and discussant feedback for each other; with student QA afterwards.

PART II

Week 9

Class 1 — Thur. October 24 (10am - 1pm) — Student Presentations

Week 10

Class 1 — Thur. October 31 (10am - 1pm) — Student Presentations

Week 11

Class 1 — Thur. November 7 (10am - 1pm) — Student Presentations

Week 12

Class 1 — Thur. November 14 (10am - 1pm) — Student Presentations

Week 13

Class 1 — Thur. November 21 (10am - 1pm) — Student Presentations